

Lab Activity 1

Uncertainties in Measurement

 40 minutes

Let's Review

Measurements always have uncertainties. It will limit you in identifying the real measurement of an object. Uncertainties in measurement are also called error; this is due to the limitations of the measuring devices. With these limitations, we cannot see or read a device beyond some fraction of the smallest division shown. The uncertainty or error indicates the maximum difference that is likely present between the measured value and the true value.

In giving the result of a measurement, we should always state the absolute uncertainty or the estimated value of the uncertainty. For example, the width of a notebook could be written as 8.8 ± 0.1 cm. The ± 0.1 cm is the absolute uncertainty in the measurement, meaning the actual width of the notebook lies between 8.7 cm and 8.9 cm. We also have what we call the percentage uncertainty, which is the ratio of the absolute uncertainty to the measured value, multiplied by 100.



Objectives

At the end of this laboratory activity, the students should be able to do the following:

- Determine the uncertainties in measurement.
- Develop skills in using measurement instruments.



Materials

- | | |
|-----------------|-----------------------|
| • block of wood | • ruler |
| • marble | • vernier caliper |
| • stone | • triple beam balance |
| • book | |



Procedure

1. There are four tables labeled Station 1, Station 2, Station 3, and Station 4. Those stations have one assigned object each, wherein you need to measure its length, width, height, and mass.
2. In station 1 you have the block of wood, in station 2 you will measure the marble, station 3 you will measure the book, and in station 4 you will measure the stone. Make sure to record your data in **Table 1.1** for the block of wood, **Table 1.2** for the marble, **Table 1.3** for the book, and **Table 1.4** for the stone.
3. Repeat the procedure three times.
4. After measuring and recording the length, width, height, and mass of each of the objects, find the density of each object in g/cm^3 . Recorded the data in **Table 1.5**.
5. Determine the percentage uncertainty in measurement of each object and record it in **Table 1.1** up to **Table 1.4**.



Reminder

To get the uncertainty in measurement follow these instructions:

- Determine the **mean** of the measurements conducted.
- Get the **range** of the highest measurement and the lowest measurement through subtraction.
- From that, calculate the **absolute uncertainty** by dividing the range into 2. You will have **mean $\pm$ absolute uncertainty**.
- Divide the absolute uncertainty by the mean, then multiply it by 100, you will now have the **percentage uncertainty**.



Data and Results

Table 1.1. Data for the block of wood

	Trial 1	Trial 2	Trial 3	Uncertainty
Length (cm)				

Height (cm)				
Width (cm)				
Mass (g)				

Table 1.2. Data for the marble

	Trial 1	Trial 2	Trial 3	Uncertainty
Length (cm)				
Height (cm)				
Width (cm)				
Mass (g)				

Table 1.3. Data for the book

	Trial 1	Trial 2	Trial 3	Uncertainty
Length (cm)				
Height (cm)				
Width (cm)				
Mass (g)				

Table 1.4. Data for the stone

	Trial 1	Trial 2	Trial 3	Uncertainty
Length (cm)				
Height (cm)				
Width (cm)				

Mass (g)				
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Table 1.5. Data for the density of each object.

Objects	Mass (g)	Volume (cm ³)	Density (g/cm ³)
block of wood			
marble			
book			
stone			



Calculations

Use the formula $\text{uncertainty} = \frac{\text{max value} - \text{min value}}{2}$ in determining the uncertainty of the measurement.



Guide Questions

Answer the following questions.

1. Which of the measuring instruments that you have used could be considered as a faulty measuring device?

2. How will you improve the experiment that could lessen the errors in measuring objects?

3. What is the importance of determining the uncertainty in measurements in real-life situations?



Conclusion



Bibliography

Giancoli, Douglas C. *Physics: Principles with Applications* (version 7th edition). Upper Saddle River, NJ: Pearson Education Inc., 2014.

Young, Hugh D., Roger A. Freedman, A. Lewis Ford, and Hugh D. Young. *University Physics*. Boston, MA: Pearson, 2014.